

WebDAV: R - gdalUtilities



WebDAV: direct access to the maps in VRT format.

R

Load libraries, select boundary box and cell size [↗](#)

Initially you need to load the following libraries and select the bounding box of the area you are interested in:

```
library(terra)
library(gdalUtilities)

bb=c(-337500.000,1242500.000,152500.000,527500.000) # Example bounding box (homolosine) for Ghana
igh='+proj=igh +lat_0=0 +lon_0=0 +datum=WGS84 +units=m +no_defs' # proj string for Homolosine projection

sg_url="/vsicurl?max_retry=3&retry_delay=1&list_dir=no&url=https://files.isric.org/soilgrids/latest/data/"
```

To a geotiff in Homolosine

This GDAL command will create a local geotiff in the Homolosine projection

```
gdal_translate(paste0(sg_url,'ocs/ocs_0-30cm_mean.vrt'),
               "./crop_roi_igh_r.tif",
               tr=c(250,250),
               projwin=bb,
               projwin_srs =igh,
               co=c("COMPRESS=DEFLATE"))
```

To a geotiff in a different projection

The following commands describe a workflow to obtain a VRT or a GeoTiff for an area of interest in a projection of your choice. In this example we will use EPSG=4326

To local VRT in homolosine (directly from the webdav connection)

The first step is to obtain a VRT for the area of interest in the Homolosine projection. We suggest to use VRT for the intermediate steps to save space and computation times.

```
gdal_translate(paste0(sg_url,'ocs/ocs_0-30cm_mean.vrt'),
               "./crop_roi_igh_r.vrt",
               of="VRT",tr=c(250,250),
               projwin=bb,
               projwin_srs =igh)
```

To a VRT in, for example, LatLong

The following command will generate a VRT in the projection of your choice:

```
gdalwarp("./crop_roi_igh_r.vrt",
         "./crop_roi_ll_r.vrt",
         s_srs=igh,
         t_srs="EPSG:4326",
         overwrite = TRUE)
```

To a final Geotiff

The following command will generate a Geotiff in the projection of your choice for the area of interest defined above

```
gdal_translate("./crop_roi_ll_r.vrt",
              "./crop_roi_ll_r.tif",
              co=c("TILED=YES", "COMPRESS=DEFLATE", "PREDICTOR=2", "BIGTIFF=YES"))
```

Read in R

Finally you can read any of the generated VRTs or GeoTiffs in R for further analysis

```
r=rast("./crop_roi_igh_r.vrt") # Or any other of the files produce above
```

To download a global geotiff in Homolosine projection

```
gdal_translate(paste0(sg_url, 'ocs/ocs_0-30cm_mean.vrt'),
              "./crop_roi_igh_r.tif",
              co=c("TILED=YES", "COMPRESS=DEFLATE", "PREDICTOR=2", "BIGTIFF=YES"))
```

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